Best Response and Nash Equilibrium

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Best Response: What is it?

- This is an extremely important concept, and is at the base of all game theory.
- The idea is quite simple, what is the best thing I can do, given what my rivals are doing.
- The most important thing here is **given what my rivals are doing**.
- Let's try the math representation.

Best Response Formalized

$$v_i\left(s_i^*, s_{-i}\right) \geq v_i\left(s_i, s_{-i}\right) \forall s_i \in S_i$$

- Easy to write for 2 players. Consider $i \in \{1, 2\}$.

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$$v_1\left(s_1^*, s_2\right) \geq v_i\left(s_1, s_2\right) \forall s_1 \in S_1, s_1^* \neq s_1$$

Prisoner's Dilemma

- This is a simple game, but has far reaching implications.

		Player 2	
		Confess	Deny
Player 1	Confess	-5,-5	0,-7
	Deny	-7,0	-1,-1

Best Responses

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Best Responses for Player 1.

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Best Responses for Player 1.

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Best Responses for Player 2.

How to write this

- $BR_1(Confess) = Confess.$
- $BR_1(Deny) = Confess.$
- $BR_2(Confess) = Confess.$
- $BR_2(Deny) = Confess.$

Let us try another one

- Simultaneous Game with two players i and j.
- $S_i = [0, 1], S_j = \{3, 5\}$.
- $v_i(s_i, s_j) = 7s_i^2 - s_j$; $v_j(s_i, s_j) = s_j - 0.5s_i$.

What is a Nash Equilibrium?

- A strategy profile $(s_1^*, s_2^* .... s_I^*)$ is a Nash Equilibrium if every player's strategy is a best response to every opponent's strategy, i.e.,

  $v_i\left(s_i^*, s_{-i}\right) \geq v_i\left(s_i, s_{-i}\right) \forall s_i \in S_i$ for every player i.

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Nash Equilibrium in Prisoner's Dilemma.

Battle of the Sexes

		Husband	
		Football	Tennis
Wife	Football	3,1	0,0
	Tennis	0,0	1,3

Game of Chicken

		Player 2	
		Swerve	Straight
Player 1	Swerve	0,0	-1,1
	Straight	1,-1	-2,-2

Penalty Shootout

		Goalkeeper	
		Left	Right
Kicker	Left	-1,1	1,-1
	Right	1,-1	-1,1